FOLD Progress

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New Baseline

Phonemic Similarity

- normalized Levenshtein distance of words based on ASJP
- words are written in a standard orthography

Likelihood `compute_overlaps` (600, 35, 35)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.143	0.419	-0.607	0.000	34
lex_sim	-0.801	0.000	-0.861	0.000	56
mut_int	-0.017	0.902	0.049	0.728	52
pho_sim	-0.656	0.000	-0.473	0.000	496

Likelihood `kl_divergence_matrix`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.111	0.532	0.677	0.000	34
lex_sim	0.800	0.000	0.830	0.000	56
mut_int	-0.010	0.944	-0.114	0.422	52
pho_sim	0.407	0.000	0.486	0.000	496

Likelihood `mae_matrix`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.143	0.419	0.607	0.000	34
lex_sim	0.801	0.000	0.861	0.000	56
mut_int	0.017	0.902	-0.049	0.728	52
pho_sim	0.656	0.000	0.473	0.000	496

Embeddings `compute_overlaps`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.078	0.663	-0.539	0.001	34
lex_sim	-0.799	0.000	-0.921	0.000	56
mut_int	0.491	0.000	0.515	0.000	52
pho_sim	-0.855	0.000	-0.681	0.000	496

Embeddings `kl_divergence_matrix`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.176	0.319	0.510	0.002	34
lex_sim	0.801	0.000	0.899	0.000	56
mut_int	-0.524	0.000	-0.533	0.000	52
pho_sim	0.876	0.000	0.652	0.000	496

Embeddings `mae_matrix`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.158	0.371	-0.500	0.003	34
lex_sim	-0.801	0.000	-0.917	0.000	56
mut_int	0.512	0.000	0.501	0.000	52
pho_sim	-0.876	0.000	-0.650	0.000	496

Next Steps

- compare embeddings across layers
- Replace parts of speech with placeholder, analyze how the spectra changes
- Define frequency bands, limit spectra by band and compare to see which bands are most discriminative
- use wals to define groups of languages that share a feature, compare spectra between groups to see if/where the feature information is stored